

Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models have done surprisingly well at answering many scientific questions. However, large language models also continue to "hallucinate" and reason incorrectly. One potential solution to these problems is multi-agent debate. Although a correct minority may exist, a confident but incorrect majority can drag this correct minority agent down and lead the group to select the wrong answer. This is known as the "sycophantic consensus." Currently, there is no mechanism within any debate system to check an individual agent's assertions against outside scientific evidence before allowing them to gain influence in the debate. Instead of using a simple vote-weighted approach, we intend to assign a trust score to every participating agent based on how much scientific evidence supports their responses. Every time agents respond during a round, the system breaks down their responses into small units of factually verifiable statements ("claims"). After breaking down the response, the system locates all relevant scientific literature that could support or contradict each claim. Because unsupported claims eventually lose influence and because the trust scores are kept in bounds, the probability increases that a correct minority will survive long enough to ultimately win. Our proposed framework produced fewer hallucinated answers than competing frameworks because all surviving claims are linked back to their originating passages of scientific evidence. To test this hypothesis, we compare our proposed framework to majority voting and other baseline systems. Once all models have finished responding to a given question, the system returns both the selected model and supporting citations for that model. The performance of our proposed framework is evaluated

via four metrics: answer accuracy, consensus collapse, minority preservation, and evidence calibration.

Keywords: Multi-agent debate, Large language models, Sycophancy, Trust calibration, Retrieval-augmented generation, Scientific question answering

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